

ASHRITH SAMBARAJU

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CAREER OBJECTIVE

Results-driven AI/ML Engineer with hands-on experience building and deploying production-grade intelligent systems in domains like energy, and e-commerce. Skilled in deep learning, computer vision, NLP, and cloud deployment (AWS, Docker), with a published research paper and internship experience solving real-world problems. Seeking a challenging role where I can apply cutting-edge AI to build scalable, impactful solutions.

EDUCATION

SPHOORTHY ENGINEERING COLLEGE (JNTUH) - 7.93/10.0	Hyderabad, TG, INDIA
<i>Bachelors of Technology in Computer Science - AIML</i>	2021-2025
LITTLE FLOWER JUNIOR COLLEGE - 8.28/10.0	Hyderabad, TG, INDIA
<i>Board of Intermediate Education - MPC</i>	2019-2021
ST. MARY'S HIGH SCHOOL - 9.0/10.0	Hyderabad, TG, INDIA
<i>Board of Secondary Education - SSC</i>	2018-2019

PUBLICATIONS

Lead Author - "*Automated Brain Tumor Classification Using Hybrid Deep Learning Models*", published in the *International Research Journal of Advanced Engineering Hub (IRJAEH)*. DOI: [10.47392/IRJAEH.2025.0318](https://doi.org/10.47392/IRJAEH.2025.0318)

EXPERIENCE

- Bharat Smart Services (Coral Innovations Pvt Ltd) - AI & ML Engineer** *September 2025 - Present*
 - ✓ Engineered a production-grade **OCR** pipeline for automated electricity bill extraction using **Google Cloud Vision**, deterministic parsing, and containerized **AWS Lambda**, supporting all DISCOMs and multiple bill formats across Telangana with **91% extraction accuracy**. Built an **insights engine** for tariff classification, electricity consumption analysis, yearly loss estimation, and consumer savings forecasting.
 - ✓ Architected and deployed an end-to-end smart meter classification system for major DISCOMs (**TPDDL, TPCODL, and KESCO**), training a **PyTorch ResNet-18 model** on close to **100 classes** of meters to achieve **92% accuracy**. Integrated OpenCV-based image quality filters (darkness/glare/blur detection) and designed a Streamlit dashboard, hosting the pipeline on **AWS Lambda via Docker and ECR** for low-latency automated top-3 predictions.
 - ✓ Built Transformer Sentinel, an AI-based predictive maintenance system for power transformers (Hack-AP, PS-3), designed for Andhra Pradesh DISCOM scenarios, achieving **86% prediction accuracy** with **XGBoost**, supported by a custom THI engine, **SHAP explainability**, and **Fast API-Streamlit** delivery.
 - ✓ Developed an AI-driven **electricity theft detection system** using 5-second smart-meter data, combining hybrid anomaly modelling, spike/dip event analytics, achieving **83% detection accuracy**. Built a full end-to-end pipeline with **Isolation Forest, LSTM Autoencoder, rule-based cue extraction**, and an interactive Streamlit dashboard for real-time monitoring, alerts, and forensic analysis.
 - ✓ Implemented a **Streamlit**-based data management dashboard with **MongoDB GridFS** for storing meter records, image handling, and automated PDF report generation.
- NextGen Edunet Foundation X EY - MERN Intern** *March 2025 - April 2025*
 - ✓ Built a dynamic e-commerce web application using the **MERN** stack as part of the Internship under Edunet Foundation in collaboration with EY, focusing on seamless user experience, user authentication, and efficient product management.
 - ✓ Structured the application around a **RESTful API**, ensuring the React frontend communicated efficiently and reliably with the MongoDB backend.
 - ✓ **Applied knowledge of secure practices** by integrating **JWT (JSON Web Tokens)** for robust user authentication and authorization, protecting session data.
 - ✓ **Demonstrated foundational MERN stack knowledge** by successfully deploying and integrating all layers of the application.
- International Institute of Information Technology (IIIT-H X Swecha) - AI Developer Intern** *May 2024 - June 2024*
 - ✓ As an AI Developer Intern, actively participated in an internship program with a 5-day workshop at **IIIT Hyderabad**, gaining hands-on experience in AI/ML, Deep Learning, & Gen AI. Developed AI-driven solutions for cultural preservation, explored

PROJECTS

- **AI-Powered Brain Tumor Detection& Treatment Recommendation System:** [Link](#)
Built a hybrid deep learning model using **VGG16**, **ResNet-50**, and **EfficientNetB2** with **Grad-CAM** for automated brain tumor detection. Integrated treatment suggestions, symptom-based screening, and PDF report generation to support clinical decisions.
- **E-Commerce Web Application (MERN Stack):** [Link](#)
Developed full-stack e-commerce web application using the **MERN** stack with product listing, authentication, and cart features. Built responsive interfaces with React, implemented **REST APIs**, and managed data handling with **MongoDB**.
- **Blockchain-Based Secure Academic Credential Management System:** [Link](#)
Designed a secure blockchain system using **Ethers** and **OTP authentication** for tamper-proof credential issuance, storage, and verification. Ensures transparency and integrity for universities, students, and verifiers

SKILLS

Programming Languages	- Python Programming, R.
AI&ML Frameworks	- NumPy, Pandas, Matplotlib, YOLO, OpenCV, LLM's, PyTorch, Streamlit.
Database& Data Storage	- MySQL, MongoDB.
IDEs	- Visual Studio, Jupyter, Google Colab
Cloud Technologies	- AWS (Lambda, ECR, API Gateway), Docker
Frontend Development	- HTML, CSS, React, Bootstrap.
Soft Skills	- Communication, Networking, Problem Solving, Prioritization & Task Management.

CERTIFICATIONS

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| • Certificate of Participation (<i>MERN Intern</i>) | - EY X Edunet Foundation |
| • Certificate of Participation (<i>AI Intern</i>) | - Summer of AI, Swecha, IIITH |
| • Introduction to Natural Language Processing | - Great Learning |
| • Developer Job Simulation | - Accenture |
| • SQL - Basic, Intermediate | - HackerRank |
| • AWS Essentials: A Complete Beginner's Guide | - Udemy |

CO-CURRICULAR ACTIVITIES

- **First Prize PRAZASTI-2K24 Techfest Hackathon** - For developing a **Robo Code** project using Java as a Team, held at our college showcasing **teamwork, problem-solving, adaptability** while innovating in AI-driven automation and programming.
- Hackathon on "Deep Dive into CNN's and NLP"- Developed a **brain tumor detection chatbot** using **APIs** and **Telegram's BotFather**, leveraging **CNN and NLP** techniques.